

Standard deviation and variance

The standard deviation and variance are most superior and widely used measure of dispersion.

Standard deviation is defined as the square root of the mean of the squared deviation from mean. It is denoted by σ .

If we have two or more comparable distributions with identical or nearly identical mean, then the distribution with smallest standard deviation has most representative mean. Hence standard deviation is extremely useful in judging the representativeness of mean.

Difference between Mean deviation and Standard deviation:-

- (1) Algebraic signs are ignored while calculating mean deviation, whereas in calculation of standard deviation signs are taken into account.
- (2) Mean deviation can be computed either from mean or median. The standard deviation, on the other hand is always computed from the arithmetic mean because sum of squares of the deviations of item from arithmetic mean is least.

$$\text{Variance} = (\text{Standard deviation})^2$$

$$\boxed{\text{Variance} = \sigma^2}$$

Calculation of Standard deviation:-

For individual Observations:-

Two methods:-

(i) deviation taken from the actual mean

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (X_i - \bar{X})^2}{N}}$$

$$\text{Say } X_i - \bar{X} = x_i$$

$$\text{ie } \sigma = \sqrt{\frac{\sum x_i^2}{N}}$$

(2) deviation taken from assumed mean:-

when the actual mean is in fraction

Say 123.674, it would be too complicated to take deviation from it and obtain the square of it. In this case we can assume mean and find S.D.

$$\sigma = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

where $d = X - A$, A is assumed mean

For Discrete Series

(1) Actual Mean method

$$\sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$$

$$\text{or } \sigma = \sqrt{\frac{\sum f x^2}{N}}$$

when $x = x_i - \bar{x}$

(2) If actual mean is in fraction we can use assumed mean method

Let A be assumed mean

find $d_i = x_i - A$

$$\text{then } \sigma = \sqrt{\frac{\sum f d_i^2}{N} - \left(\frac{\sum f d_i}{N}\right)^2}$$

For Continuous Series:-

For continuous series any of the above method can be used. In practice step deviation method is used, formula is given below

$$\sigma = \sqrt{\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N}\right)^2} \times h$$

where $d = \frac{m - A}{h}$, A = assumed mean
m = mid points of classes

Q1. Table below gives the marks obtained by B.com students with Roll no. 1 to 10 at an examination. Calculate the standard deviation.

Roll no.:	1	2	3	4	5	6	7	8	9	10
Marks:	43	48	65	57	31	60	37	48	78	59

Marks	$d = x - 50$	d^2
x		
43	-7	49
48	-2	4 4
65	15	225
57	7	49
31	-19	361
60	10	100
37	-13	169
48	-2	4
78	28	784
59	9	81
$\Sigma x = 526$	$\Sigma d = 26$	$\Sigma d^2 = 1826$

$$N = 10, \Sigma x = 526$$

let assumed mean $A = 50$

$$d = x - A = x - 50.$$

$$\sigma = \sqrt{\frac{\Sigma d^2}{N} - \left(\frac{\Sigma d}{N}\right)^2}$$

$$= \sqrt{\frac{1826}{10} - \left(\frac{26}{10}\right)^2} = \sqrt{182.6 - (2.6)^2}$$

$$= \sqrt{175.84}$$

$$\sigma = 13.26$$

Q Calculate the S.D of following observations
 240.12, 240.13, 240.15, 240.12, 240.15, 240.17,
 240.16, 240.22, 240.12

Ans let $A = 240$

$$\sigma = 0.033$$

Q2 Calculate S.D from the following data.

Size of item: 3.5 4.5 5.5 6.5 7.5 8.5 9.5
 frequency: 3 7 22 60 85 32 8

Soln

X	f	$d = X - 6.5$	fd	fd^2
3.5	3	-3	-9	27
4.5	7	-2	-14	28
5.5	22	-1	-22	22
6.5	60	0	0	0
7.5	85	1	85	85
8.5	32	2	64	128
9.5	8	3	24	72
$\Sigma f = 217$			$\Sigma fd = 128$	$\Sigma fd^2 = 362$

let $A = 6.5$

$$d = X - 6.5$$

$$N = 217$$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2}$$

$$= \sqrt{\frac{362}{217} - \left(\frac{128}{217}\right)^2}$$

$$\sigma = \sqrt{1.668 - .349} = 1.149$$

$$\sigma = 1.149$$

Q:- calculate S.D for the following data:-

Profits (000's)	20-30	30-40	40-50	50-60	60-70	70-80
No. of Cos.	30	58	62	85	112	70
	80-90	90-100				
	57	26				

X	m	f	$d = \frac{m-55}{10}$	d^2	fd	fd^2
20-30	25	30	-3	9	-90	270
30-40	35	58	-2	4	-116	232
40-50	45	62	-1	1	-62	62
50-60	55	85	0	0	0	0
60-70	65	112	+1	1	112	112
70-80	75	70	+2	4	140	280
80-90	85	57	+3	9	171	513
90-100	95	26	+4	16	104	416
					259	1885

$N = 500$

Let assumed mean = 55

$$d = \frac{m - A}{h}$$

$$d = \frac{m - 55}{10}$$

$\sum fd = 259$ $\sum fd^2 = 1885$

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2 \times h} \\ &= \sqrt{\frac{1885}{500} - \left(\frac{259}{500}\right)^2 \times 10} \\ &= \sqrt{3.77 - .268 \times 10} \end{aligned}$$

$\sigma = 18.71$

Q Find S.D for the following distribution

Age	20-25	25-30	30-35	35-40	40-45	45-50
N ^o of Person	170	110	80	45	40	35

Ans take $A = 32.5$

$$\sigma = 7.935$$

Coefficient of variation :-

standard deviation discussed above is an absolute measure of dispersion. The corresponding relative measure is coeff. of variation

It is used to compare the variability of two or more series. The series for which the coeff. of variation is greater is said to be more variable or less consistent, for which the coeff. of variation is less is said to be less variable or more consistent, more stable or more uniform.

Coeff. of variation is denoted by

$$C.V = \frac{\sigma}{\text{mean}} \times 100$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

Note:- When two series have different unit of measurements, or their mean differs, the CV should be used as measure of dispersion.

Q Two score of two players A and B in ten innings during a certain season are :-

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

For which of two players A, B is more consistent in Scoring.

Soln we will find coeff of variation of two series

Series A:-

X	$d = x - \bar{x}$	$(x - \bar{x})^2$
32	-14	196
28	-18	324
47	1	1
63	17	289
71	25	625
39	-7	49
10	-36	1296
60	14	196
96	50	2500
14	-32	1024
$\Sigma x = 460$	$\Sigma d = 0$	$\Sigma d^2 = 6500$

$$\bar{x} = \frac{\Sigma x}{N} = \frac{460}{10} = 46$$

~~$$\sigma = \sqrt{\frac{\Sigma d^2}{N}} = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}}$$~~

$$\sigma_A = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}}$$

$$\sigma_A = \sqrt{\frac{6500}{10}}$$

$$\sigma_A = 25.5$$

slly	y	y - $\bar{y}$	(y - $\bar{y}$) ²
	19	-31	961
	31	-19	361
	48	-2	4
	53	3	9
	67	17	289
	90	40	1600
	10	-40	1600
	62	+12	144
	40	-10	100
	80	30	900
$\Sigma y =$	500	0	5968

$$\Sigma y = 500$$

$$\bar{y} = \frac{\Sigma y}{10} = 50$$

$$\sigma_B = \sqrt{\frac{\Sigma (y - \bar{y})^2}{N}} = 24.43$$

$$\boxed{\sigma_B = 24.43}$$

$$C_o.V(A) = \frac{\sigma}{\bar{y}} \times 100 = 55.43$$

$$C_o.V(B) = \frac{\sigma}{\bar{y}} \times 100 = 48.86$$

as C.o.V of players B is less i.e. Players B are consistent in scoring.

Q - Following are the marks obtained by two students A and B in 10 sets of examination

Sets: 1 2 3 4 5 6 7 8 9 10

A's: 44 80 76 48 52 72 68 56 60 64

B's: 48 75 54 60 63 69 72 51 57 56

If the consistency of performance is the criteria of awarding the prize, who should get the prize.

Ans

$$\sigma(A) = 11.489$$

$$\sigma(B) = 8.617$$

$$C.V(A) = 18.53\% \quad C.V(B) = 14.24\%$$

ie student B is more consistent because
Coff. of variation is less than that of A
ie B should get the prize.

Q:- Suppose samples of polythene bags from two manufacturers A and B, are tested by a prospective buyer for bursting pressure with the following results:-

Bursting Pressure (lb)	No. of Bags	
	A	B
5-9.9	2	9
10.0-14.9	9	11
15.0-19.9	29	18
20.0-24.9	54	32
25.0-29.9	11	27
30.0-34.9	5	13

(a) which set of bags has highest average bursting pressure.

(b) which has more uniform pressure? if prices are same

(c) which manufacturer's bag would be preferred by the buyer? why?

converting the class intervals in to cont class

$$\frac{10-9.9}{2} = \frac{.1}{2} = .05$$

ie each class interval $a-b$ becomes

$$(a-.05) - (b+.05)$$

For A	m	No. of Bags	d	fd	fd ²
4.95 - 9.95	7.45	2	-2	-4	16
9.95 - 14.95	12.45	9	-1	-9	81
14.95 - 19.95	17.45	29	0	0	0
19.95 - 24.95	22.45	54	1	54	54
24.95 - 29.95	27.45	11	2	22	44
29.95 - 34.95	32.45	5	3	15	45

$$N = 110$$

Let assumed mean = 17.45

$$h = 5$$

Using Step deviation method

$$d = \frac{m - 17.45}{5}$$

$$\sum fd = 78$$

$$\sum fd^2 = 160$$

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h$$

$$\sigma_A = \sqrt{\frac{160}{110} - \left(\frac{78}{110}\right)^2} \times 5 = 4.88$$

$$C.V = \frac{\sigma}{\bar{X}_A} \times 100$$

$$\text{Now } \bar{X} = A + \frac{\sum fd}{N} \times h$$

$$\bar{X} = 17.45 + \frac{78}{110} \times 5 = 21$$

ie $C.V. = \frac{\sigma_A}{\bar{X}} \times 100$

$$C.V._A = 23.24\%$$

For B

Pressure	m	f	$d = \frac{m-17.45}{5}$	fd	fd ²
4.95-9.95	7.45	9	-2	-18	36
9.95-14.95	12.45	11	-1	-11	11
14.95-19.95	17.45	18	0	0	0
19.95-24.95	22.45	32	1	32	32
24.95-29.95	27.45	27	2	54	108
29.95-34.95	32.45	13	3	39	117
$\Sigma fd = 96$					$\Sigma fd^2 = 304$

$$\sigma_B = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times 5$$

$$\sigma_B = 7.075$$

$$\bar{X}_B = A + \frac{\Sigma fd}{N} \times h$$

$$\bar{X}_B = 17.45 + \frac{96}{110} \times 5 = 21.81$$

$$so \quad C.V.(B) = \frac{\sigma_B}{\bar{X}_B} \times 100$$

$$= 32.44\%$$

(a) ie $\bar{X}_A = 21$ $\bar{X}_B = 21.81$

ie Set B has highest average bursting pressure.

(b) $C.V(A) = 23.24\%$

$C.V(B) = 32.44\%$

If prices are same then C.V it is clear that Set (A) has more uniform pressure.

(c) Set (A) will be preferred.

Merits of S.D :-

- (1) It is best measures of variation as it is based on every item of distribution and is less affected by fluctuations of sampling than the other measures of dispersion.
- (2) For comparing variability of two or more distribution coeff. of variation is considered to be most appropriate and its based on mean standard deviation.
- (3) S.D is mostly used in further statistical work. For ex computing skewness, correlation etc.
- (4) It is possible to calculate the combined S.D of two or more groups. This is not possible with any other measure.

formula is $\bar{X}_{12} = \frac{N_1 \bar{X}_1 + N_2 \bar{X}_2}{N_1 + N_2}$

Q: coeff. of variation of wages of male workers and female workers are 55% & 70% resp., while S.D are 22.0 and 15.4 resp. calculate overall average wages of all workers